Customer Refund Portal

PRD-REF-021 | Product Requirements Document

FICTIONAL VALIDATION DOCUMENT - contains no real customer or production data

Status	Draft	**Owner**	Product & Engineering
Version	1.0	**Updated**	15 Oct 2026
Audience	Product, QA, Engineering	**Classification**	Synthetic / Non-production

1. Purpose and scope

Provide a self-service portal where a customer can review an eligible purchase and request a refund. The portal records the request, displays the decision status, and sends a notification when the status changes. The initial release supports card and virtual-account purchases only.

2. Functional requirements

ID	Requirement	Acceptance criteria	Priority
FR-01	View eligible purchases	Show purchases within the 90-day eligibility window, newest first. Mask payment instrument references except the final four characters.	High
FR-02	Submit refund request	Require a reason code and optional description up to 500 characters. Prevent duplicate open requests for the same purchase.	High
FR-03	Validate amount	Requested amount must be greater than zero and no greater than the remaining refundable amount.	High
FR-04	Track request	Display request ID, submitted time, current status, and the latest status-change time.	High
FR-05	Withdraw request	Allow withdrawal only while status is RECEIVED or UNDER_REVIEW. Record withdrawal time and actor.	Medium
FR-06	Notify status change	Send an in-app notification for every committed status transition. Retry transient delivery failures up to three times.	Medium
FR-07	Audit changes	Store actor, previous status, new status, event time, and correlation ID. Audit entries are append-only.	High

ID	Requirement	Acceptance criteria	Priority
FR-08	Accessibility	All actions must be keyboard accessible, have visible focus, and expose a meaningful accessible name.	Medium

3. Status and decision mapping

Code	Portal status	Meaning	System behavior
RCV	RECEIVED	Request created and awaiting eligibility check	No payout initiated
REV	UNDER_REVIEW	Eligibility passed; an agent decision is required	Customer may withdraw
APR	APPROVED	Decision accepted and payout instruction queued	Payout starts asynchronously
REJ	REJECTED	Decision rejected with a reason code	No payout initiated
PAY	PAID	Payment provider confirms refund settlement	Terminal success
ERR	PAYOUT_FAILED	Provider reports a permanent payout failure	Show support contact guidance
WDR	WITHDRAWN	Customer withdrew before approval	Terminal customer action

4. Validation and error messages

Error code	Customer message	Trigger
RF-400	Refund amount is invalid.	The amount is zero, negative, or above the refundable balance.
RF-409	A refund request is already open.	An open request exists for the same purchase.
RF-422	This purchase is no longer eligible.	The eligibility window has expired or the purchase is excluded.
RF-503	Refund service is temporarily unavailable.	The request was not submitted; the customer may retry.

5. Key user flow

1. Customer opens Purchase History and selects an eligible purchase.
2. Portal calculates the remaining refundable amount and displays the refund form.
3. Customer submits the reason and amount; the service validates the request atomically.
4. Portal displays the request ID and RECEIVED status after successful creation.
5. Later status changes appear in the portal and trigger notifications.

6. Non-functional constraints

The purchase list should render within 2 seconds for 95% of requests under 300 concurrent sessions. The create-request endpoint must be idempotent when the same idempotency key is reused. All timestamps are stored in UTC and rendered in the user profile timezone.